

WAZUH – VIRUSTOTAL INTEGRATION

Automated Threat Detection & Active Response

Phase 1: Configuring the Ubuntu Endpoint (SOC_Lab_Endpoint)

1. Enable Real-Time File Integrity Monitoring (FIM)

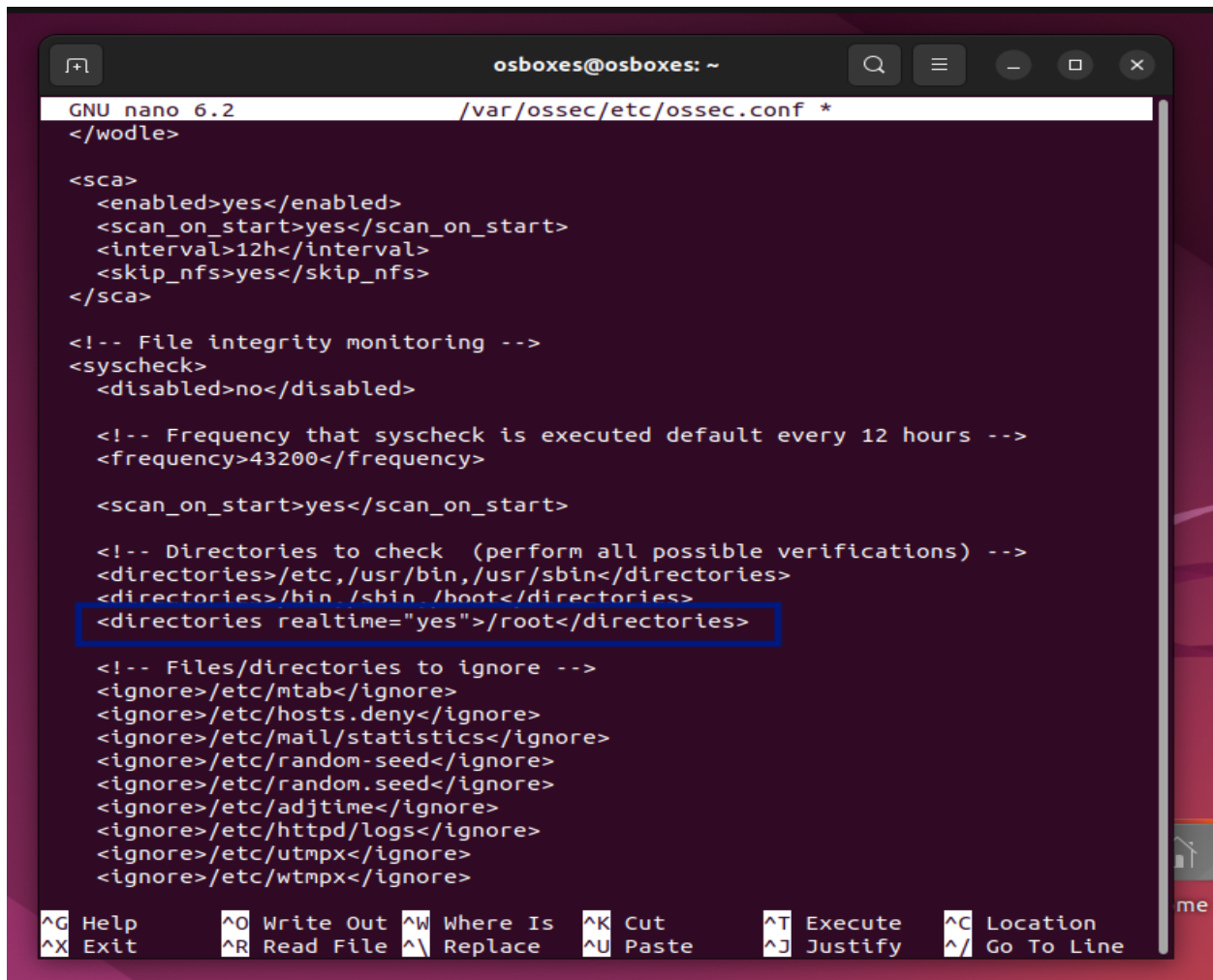
The `/root` directory needs to be monitored in real-time so that any newly downloaded file triggers an inspection.

Open the Wazuh Agent configuration file:

```
sudo nano /var/ossec/etc/ossec.conf
```

Locate the `<syscheck>` block and add this line inside it:

```
<directories realtime="yes">/root</directories>
```



```
osboxes@osboxes: ~  
GNU nano 6.2 /var/ossec/etc/ossec.conf *  
</wodle>  
  
<sca>  
  <enabled>yes</enabled>  
  <scan_on_start>yes</scan_on_start>  
  <interval>12h</interval>  
  <skip_nfs>yes</skip_nfs>  
</sca>  
  
<!-- File integrity monitoring -->  
<syscheck>  
  <disabled>no</disabled>  
  
  <!-- Frequency that syscheck is executed default every 12 hours -->  
  <frequency>43200</frequency>  
  
  <scan_on_start>yes</scan_on_start>  
  
  <!-- Directories to check (perform all possible verifications) -->  
  <directories>/etc,/usr/bin,/usr/sbin</directories>  
  <directories>/bin /sbin /boot</directories>  
  <directories realtime="yes">/root</directories>  
  
  <!-- Files/directories to ignore -->  
  <ignore>/etc/mtab</ignore>  
  <ignore>/etc/hosts.deny</ignore>  
  <ignore>/etc/mail/statistics</ignore>  
  <ignore>/etc/random-seed</ignore>  
  <ignore>/etc/random.seed</ignore>  
  <ignore>/etc/adjtime</ignore>  
  <ignore>/etc/httpd/logs</ignore>  
  <ignore>/etc/utmpx</ignore>  
  <ignore>/etc/wtmpx</ignore>  
  
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location  
^X Exit      ^R Read File ^\ Replace  ^U Paste     ^J Justify  ^_ Go To Line
```

2. Create the Custom Active Response Script

When VirusTotal flags a file, this script will run automatically to delete the threat.

Install jq:

```
Sudo apt install jq
```

Create a new script file:

```
sudo nano /var/ossec/active-response/bin/remove-threat.sh
```

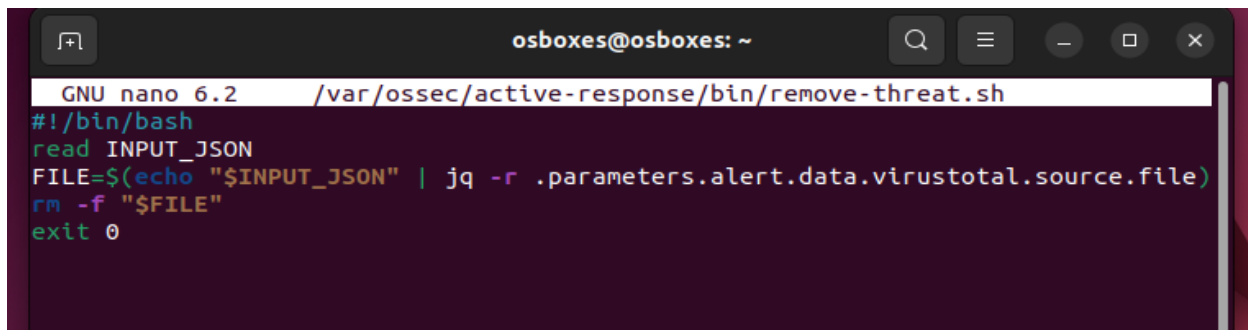
Paste the following:

```
#!/bin/bash
read INPUT_JSON
FILE=$(echo "$INPUT_JSON" | jq -r .parameters.alert.data.virustotal.source.file)
rm -f "$FILE"
exit 0
```

Save with **Ctrl O**

Press **enter**

Press **Ctrl X** to exit

A screenshot of a terminal window titled 'osboxes@osboxes: ~'. The window shows the GNU nano 6.2 editor editing the file '/var/ossec/active-response/bin/remove-threat.sh'. The script content is: #!/bin/bash, read INPUT_JSON, FILE=\$(echo "\$INPUT_JSON" | jq -r .parameters.alert.data.virustotal.source.file), rm -f "\$FILE", and exit 0. The terminal has a dark background with light-colored text.

Set permissions

```
sudo chmod 750 /var/ossec/active-response/bin/remove-threat.sh
sudo chown root:wazuh /var/ossec/active-response/bin/remove-threat.sh
```

Restart the Agent

```
sudo systemctl restart wazuh-agent
```

Phase 2: Configuring the Wazuh Server (Kali Linux Manager)

Configure the Wazuh Manager to connect with VirusTotal and trigger our custom deletion script.

1. Add Custom Detection Rules

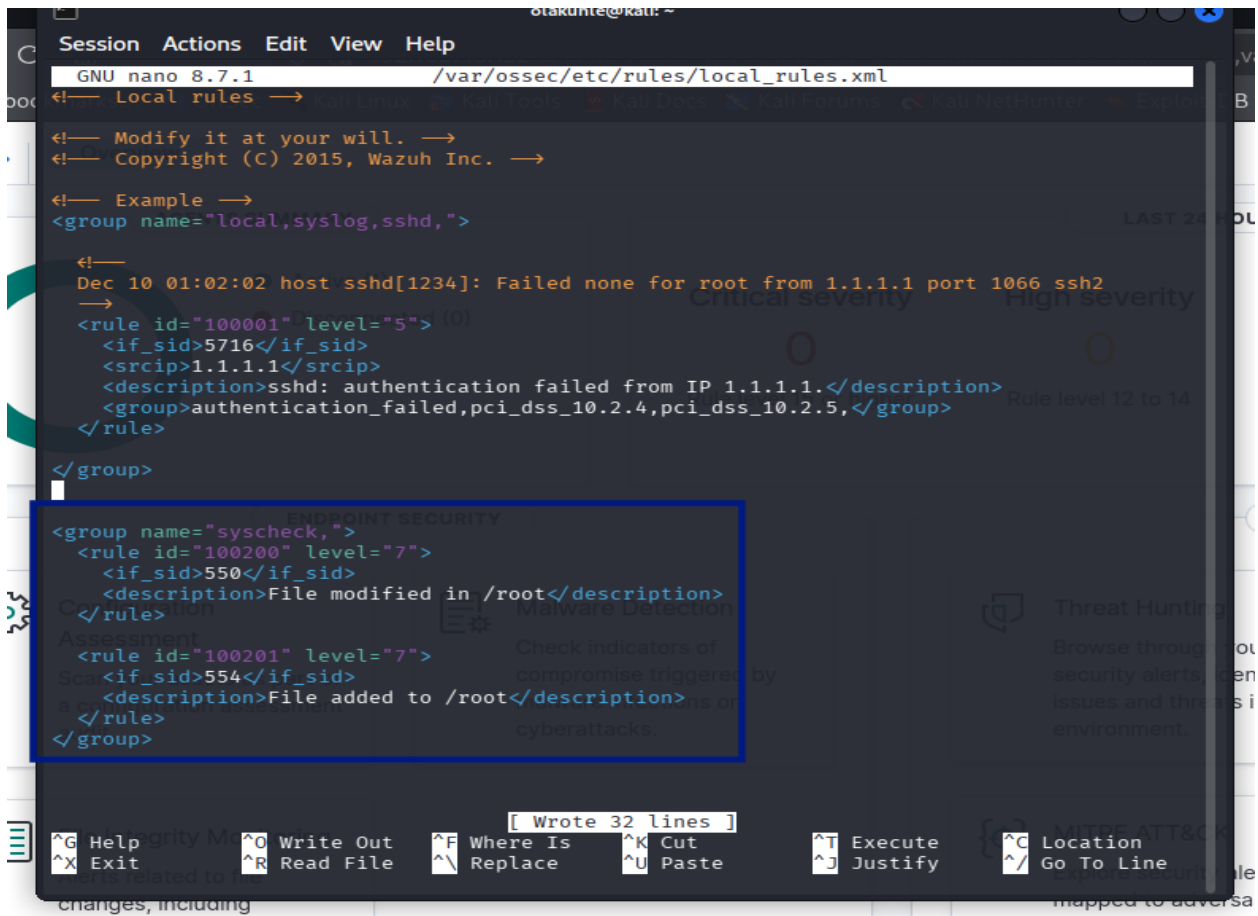
Open the local rules configuration file:

```
sudo nano /var/ossec/etc/rules/local_rules.xml
```

Add the following rules inside the <group> tags at the bottom:

```
<group name="syscheck,">
  <rule id="100200" level="7">
    <if_sid>550</if_sid>
    <description>File modified in /root directory</description>
  </rule>

  <rule id="100201" level="7">
    <if_sid>554</if_sid>
    <description>File added to /root</description>
  </rule>
</group>
```



```
GNU nano 8.7.1 /var/ossec/etc/rules/local_rules.xml
<!-- Local rules -->
<!-- Modify it at your will. -->
<!-- Copyright (C) 2015, Wazuh Inc. -->

<!-- Example -->
<group name="local,syslog,sshd,">

  <!--
  Dec 10 01:02:02 host sshd[1234]: Failed none for root from 1.1.1.1 port 1066 ssh2
  -->
  <rule id="100001" level="5"> (0)
    <if_sid>5716</if_sid>
    <srcip>1.1.1.1</srcip>
    <description>sshd: authentication failed from IP 1.1.1.1.</description>
    <group>authentication_failed,pci_dss_10.2.4,pci_dss_10.2.5,</group>
  </rule>

</group>

<group name="syscheck,">
  <rule id="100200" level="7">
    <if_sid>550</if_sid>
    <description>File modified in /root</description>
  </rule>
  <rule id="100201" level="7">
    <if_sid>554</if_sid>
    <description>File added to /root</description>
  </rule>
</group>
```

2. Configure VirusTotal and Active Response in ossec.conf

Open the Wazuh Manager main configuration file:

```
sudo nano /var/ossec/etc/ossec.conf
```

Enable VirusTotal Integration: Add this within the <integration> block:

```
<integration>
  <name>virustotal</name>
  <api_key>ADD_YOUR_VIRUSTOTAL_API_KEY_HERE</api_key>
  <rule_id>100200,100201</rule_id>
  <alert_format>json</alert_format>
</integration>
```

Open an account on Virustotal to get your API key.

100200, 100201 - Means send file hashes from rules 100200 & 100201 to Virustotal

```
Session Actions Edit View Help
GNU nano 8.7.1 /var/ossec/etc/ossec.conf *
<skip_nfs>yes</skip_nfs>

<ignore>/var/lib/containerd</ignore>
<ignore>/var/lib/docker/overlay2</ignore>
</rootcheck>

<wodle name="cis-cat">
  <disabled>yes</disabled>
  <timeout>1800</timeout>
  <interval>1d</interval>
  <scan-on-start>yes</scan-on-start>

  <java_path>wodles/java</java_path>
  <ciscat_path>wodles/ciscat</ciscat_path>
</wodle>

<!-- Osquery integration -->
<wodle name="osquery">
  <disabled>yes</disabled>
  <run_daemon>yes</run_daemon>
  <log_path>/var/log/osquery/osqueryd.results.log</log_path>
  <config_path>/etc/osquery/osquery.conf</config_path>
  <add_labels>yes</add_labels>
</wodle>

<integration>
  <name>virustotal</name>
  <api_key>09b42c5b5afc29c35ea68005f7a</api_key>
  <rule_id>100200,100201</rule_id>
  <alert_format>json</alert_format>
</integration>

<!-- System inventory -->
<wodle name="syscollector">
```

Define the command (This command exists and can be triggered):
Still inside ossec.conf, add these:

```
<command>
  <name>remove-threat</name>
  <executable>remove-threat.sh</executable>
</command>
```

Define the Active Response Policy: Trigger the command when VirusTotal alert hits rule 87105 (Malware confirmed):

```
<active-response>
  <command>remove-threat</command>
  <location>local</location>
  <rules_id>87105</rules_id>
</active-response>
```

```

GNU nano 8.7.1 /var/ossec/etc/ossec.conf *
<executable>netsh.exe</executable>
<timeout_allowed>yes</timeout_allowed>
</command>

<command>
<name>remove-threat</name>
<executable>remove-threat.sh</executable>
</command>

<active-response>
  active-response options here
</active-response>

<active-response>
<command>remove-threat</command>
<location>local</location>
<rules_id>87105</rules_id>
</active-response>

<localfile>
<log_format>command</log_format>
<command>df -P</command>
<frequency>360</frequency>
</localfile>

<localfile>
<log_format>full_command</log_format>
<command>netstat -tulnp | sed 's/\([[:alnum:]]\+\)\ \([[:digit:]]\+\) \([[:digit:]]\+\) \([[:digit:]]\+\)'
<alias>netstat listening ports</alias>
<frequency>360</frequency>
</localfile>

```

Save with **Ctrl O**
Press **enter**
Press **Ctrl X** to exit

Restart the Wazuh Manager

```
sudo systemctl restart wazuh-manager
```

Phase 3: Threat Simulation & Verification

Simulate the Attack

On the SOC_Lab_Endpoint, install curl and download an EICAR standard anti-malware test file directly into the monitored */root* directory:

Install curl

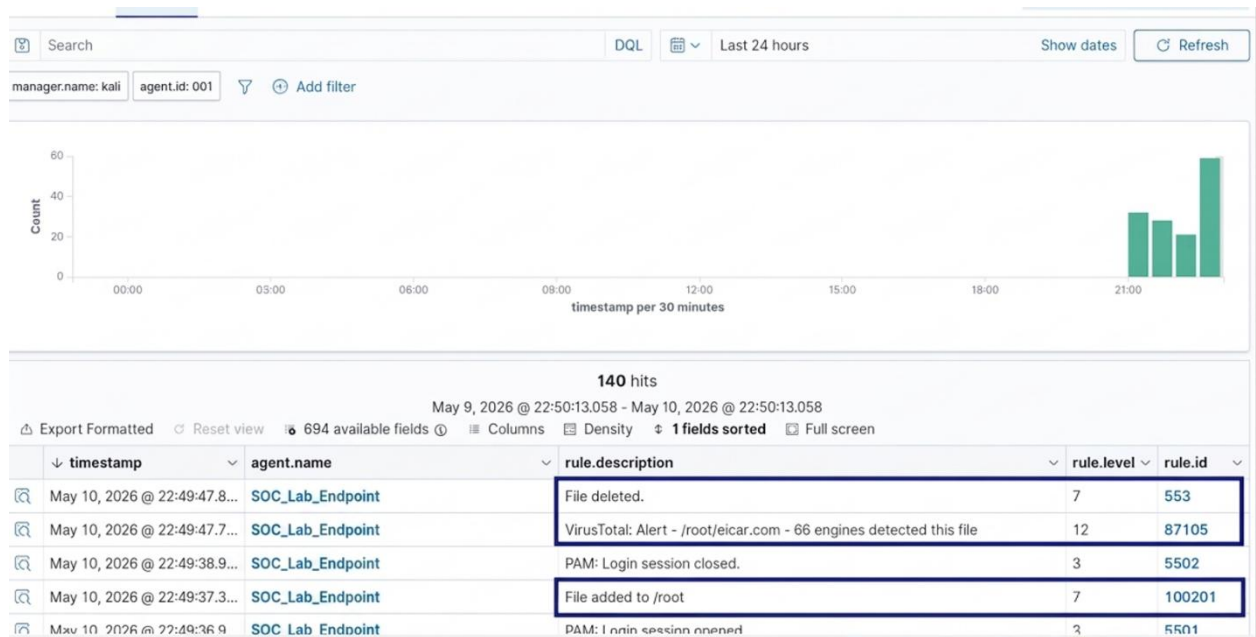
```
sudo apt install curl -y
```

Then download EICAR

```
sudo curl -Lo /root/eicar.com https://secure.eicar.org/eicar.com
```

Expected Results

- i. The file will initially land in /root/eicar.com.
- ii. Wazuh's detects it in real-time and sends the hash to VirusTotal.
- iii. VirusTotal flags it, firing Rule 87105.
- iv. The Active Response script runs instantly, and the file is deleted.



Verify completion by running `ls /root/eicar.com`—it should show No such file or directory.

```
osboxes@osboxes: ~  
osboxes@osboxes:~$ sudo curl -Lo /root/eicar.com https://secure.eicar.org/eicar.com  
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current  
           Dload  Upload   Total   Spent    Left  Speed  
100    68    100    68     0     0    71     0  --:--:-- --:--:-- --:--:--    71  
osboxes@osboxes:~$ sudo ls -la /root/eicar.com  
ls: cannot access '/root/eicar.com': No such file or directory  
osboxes@osboxes:~$
```